12 Troubleshooting your application

This chapter covers

- Understanding the nature of an exception
- Using StatusCodePages middleware to control the display of client error messages
- Implementing logging within a Razor Pages application

By now, you've probably written a fair amount of code as you have worked your way through the chapters. Chances are you have encountered an error or two while doing so. Some of these will be compile-time errors your development tool will have caught, so you can fix them as you work. Others will have surfaced only at run time, raising exceptions that result in an error page of some kind being displayed. As you build more complex applications, other types of errors creep in: logical errors. These kinds of errors won't stop your application from running, but they will result in unpredictable behavior. Coding errors, whether they are simply typos or logical errors, are a fact of a developer's life.

In this chapter, we will explore the nature of errors in a bit more detail, so that you can understand what you're dealing with. Ideally, you should code to prevent errors in the first place, but it is the nature of development that bugs will creep in and that forces outside of your control will affect things such as the availability of databases, mail servers, and other external services. We will look at how the application behaves at runtime when an error happens that results in an exception being raised and what opportunities there are to take control over that behavior.